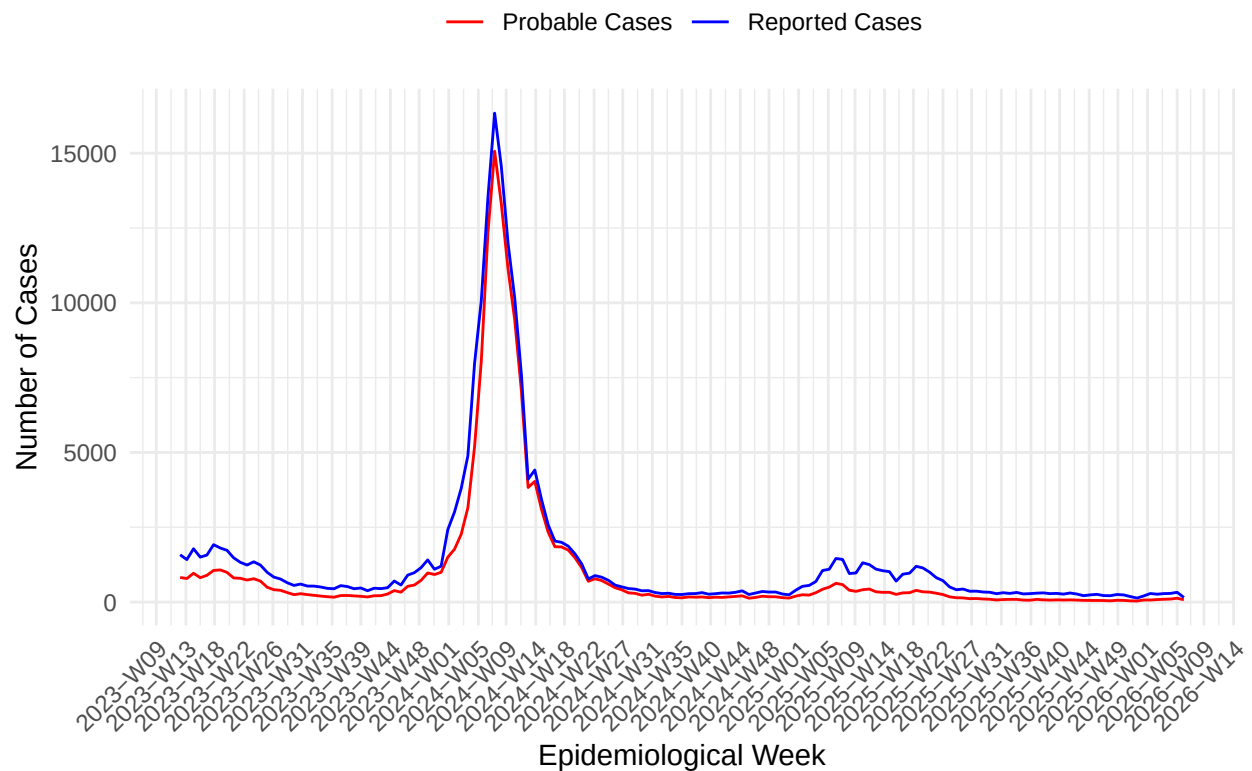


Dengue Forecasting in Rio de Janeiro

Exploratory Data Analysis

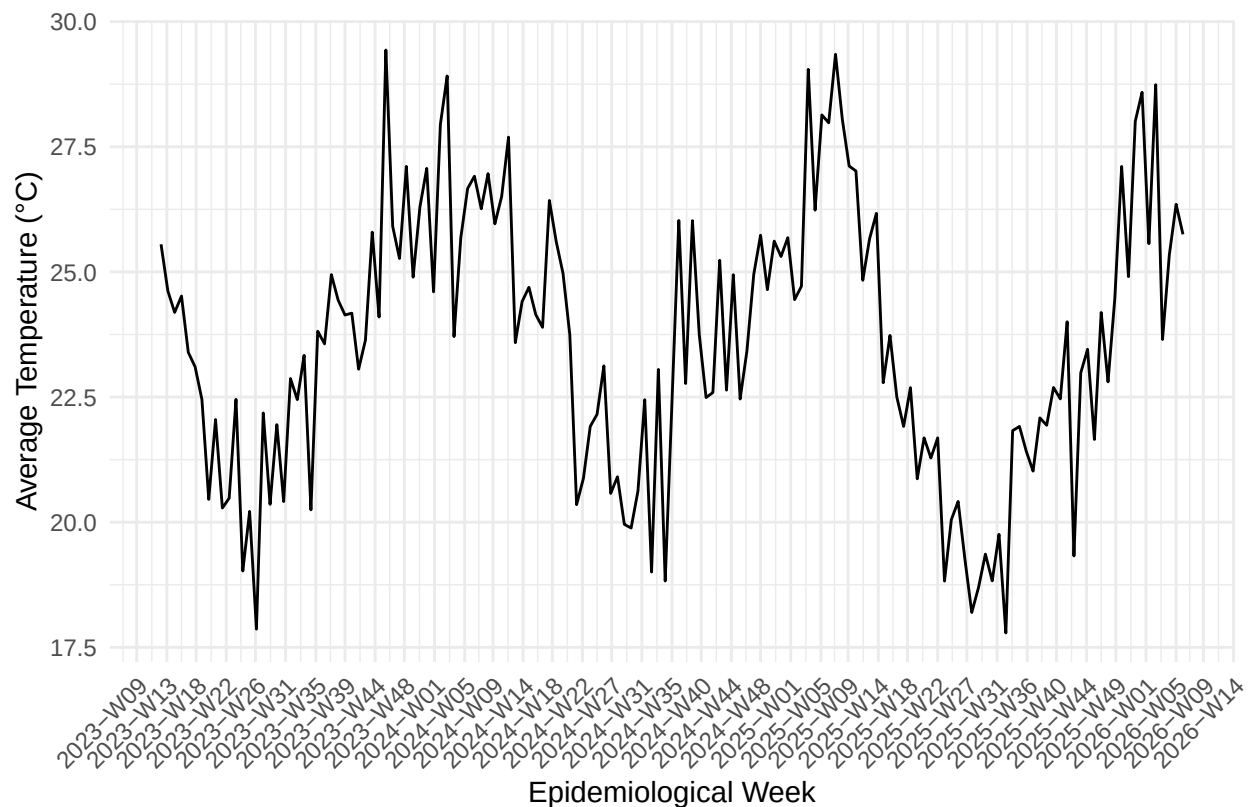
Time Series of Dengue Cases

```
ggplot() +  
  geom_line(data = dengue_climate_rj,  
            aes(x = data_iniSE, y = casprov, color = "Probable Cases")) +  
  geom_line(data = dengue_climate_rj,  
            aes(x = data_iniSE, y = casos, color = "Reported Cases")) +  
  labs(title = "", x = "Epidemiological Week", y = "Number of Cases") +  
  scale_x_date(date_labels = "%Y-W%V", date_breaks = "1 month") +  
  scale_color_manual(values = c("Reported Cases" = "blue",  
                                "Probable Cases" = "red")) +  
  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1),  
        legend.title = element_blank(),  
        legend.position = "top")
```



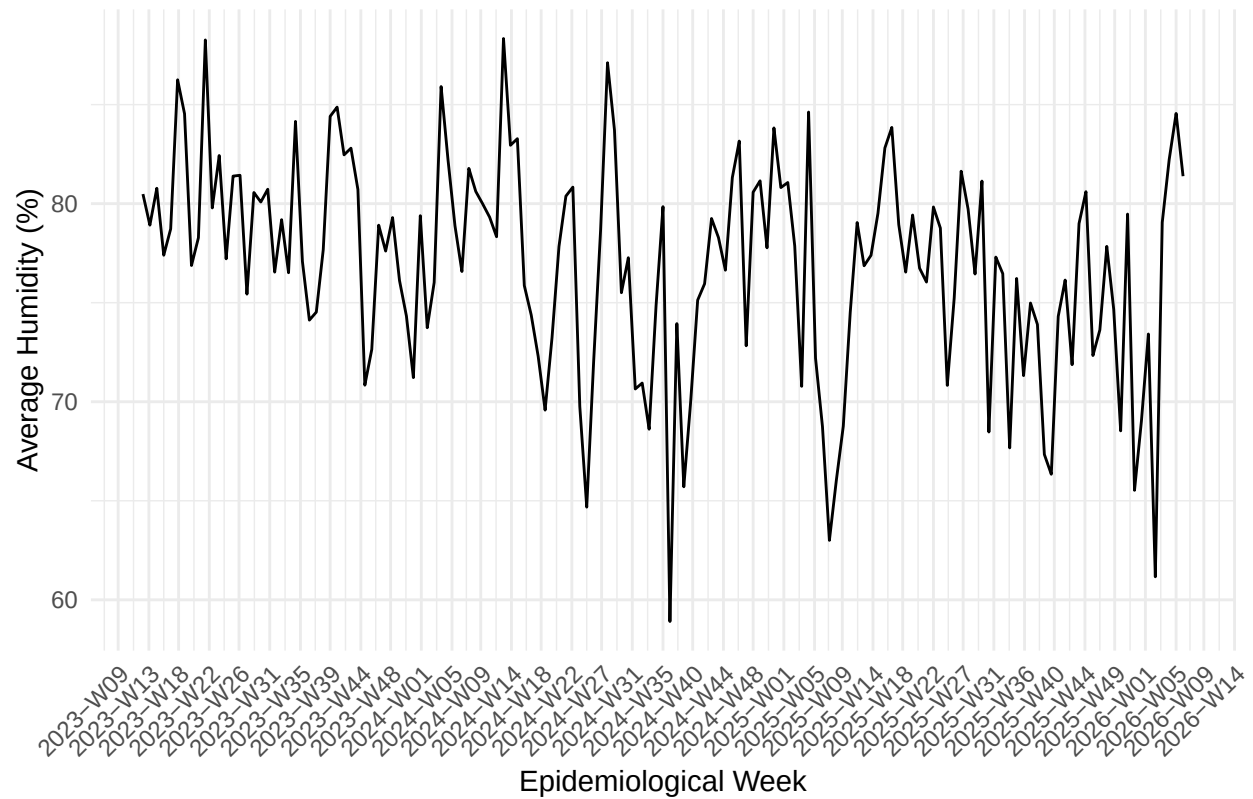
Time Series of Temperature

```
ggplot(dengue_climate_rj, aes(x = data_iniSE)) +  
  geom_line(aes(y = tempmed)) +  
  labs(title = "", x = "Epidemiological Week", y = "Average Temperature (°C)") +  
  scale_x_date(date_labels = "%Y-W%V", date_breaks = "1 month") +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1),  
        legend.title = element_blank(),  
        )
```



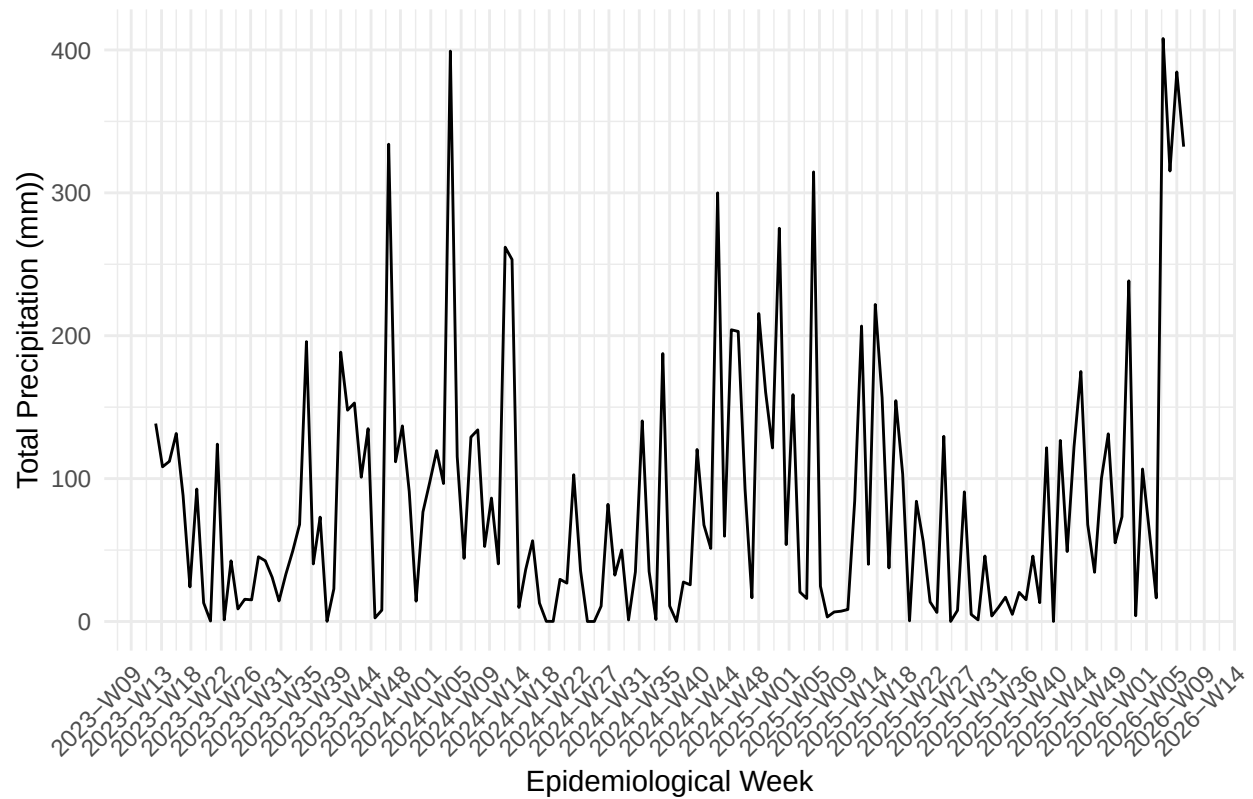
Time Series of Humidity

```
ggplot(dengue_climate_rj, aes(x = data_iniSE)) +  
  geom_line(aes(y = umidmed)) +  
  labs(title = "", x = "Epidemiological Week", y = "Average Humidity (%)") +  
  scale_x_date(date_labels = "%Y-W%V", date_breaks = "1 month") +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1),  
        legend.title = element_blank(),  
        )
```



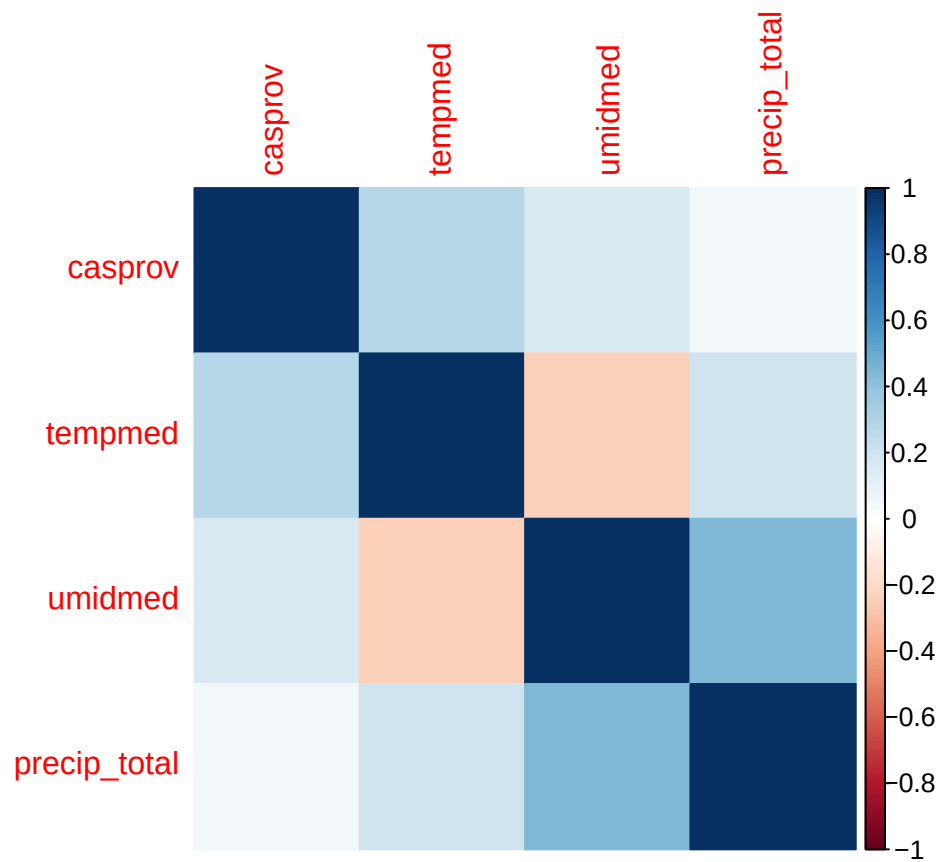
Time Series of Precipitation

```
ggplot(dengue_climate_rj, aes(x = data_iniSE)) +
  geom_line(aes(y = precip_total)) +
  labs(title = "", x = "Epidemiological Week", y = "Total Precipitation (mm)") +
  scale_x_date(date_labels = "%Y-W%V", date_breaks = "1 month") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.title = element_blank(),
        )
```



Correlation Analysis

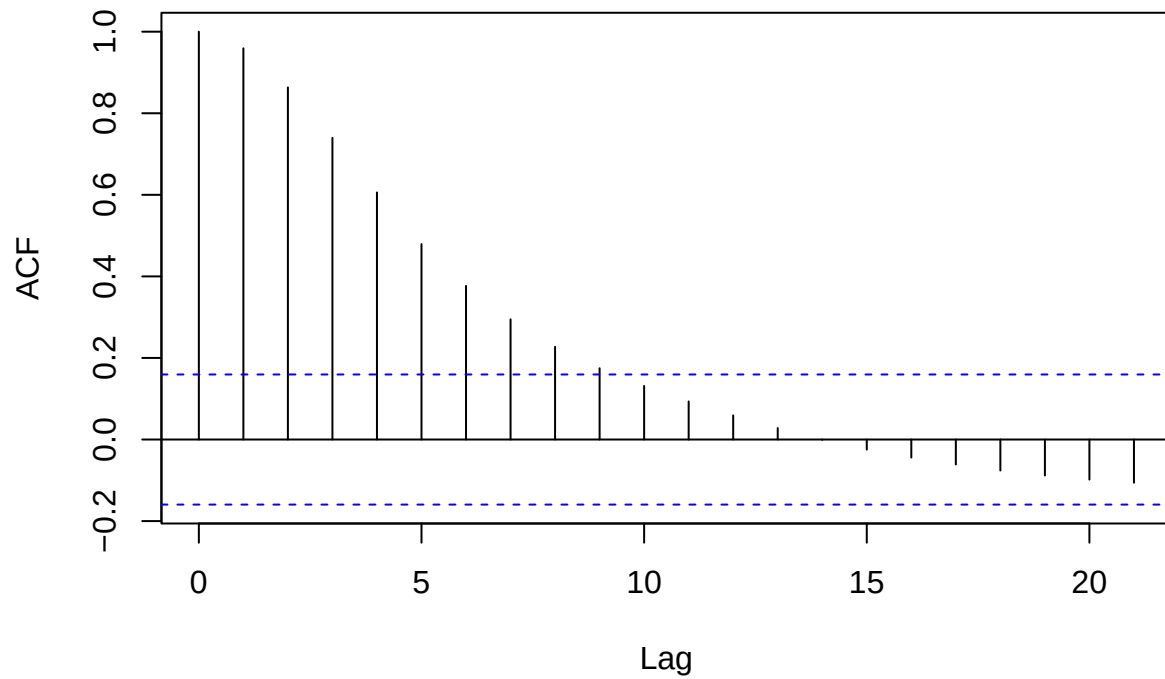
```
cor_data <- dengue_climate_rj %>%
  select(casprov, tempmed, umidmed, precip_total) %>%
  drop_na()
cor_matrix <- cor(cor_data)
corrplot::corrplot(cor_matrix, method = "color")
```



Autocorrelation Analysis

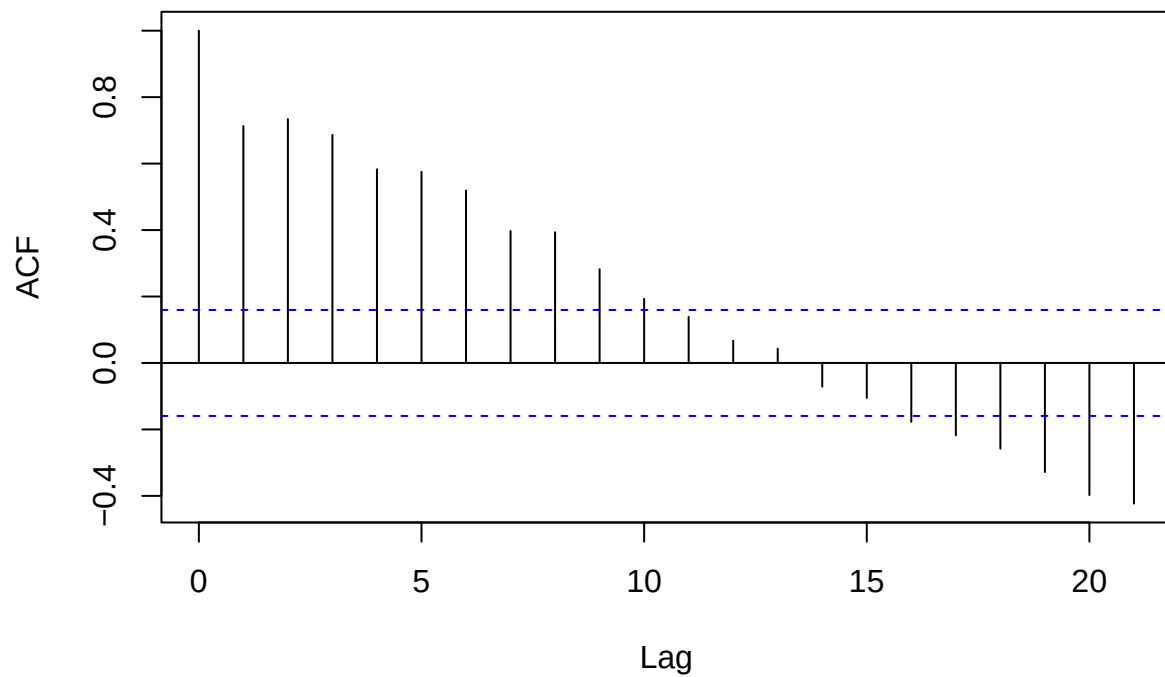
```
acf(dengue_climate_rj$casprov, main = "ACF of Dengue Cases")
```

ACF of Dengue Cases



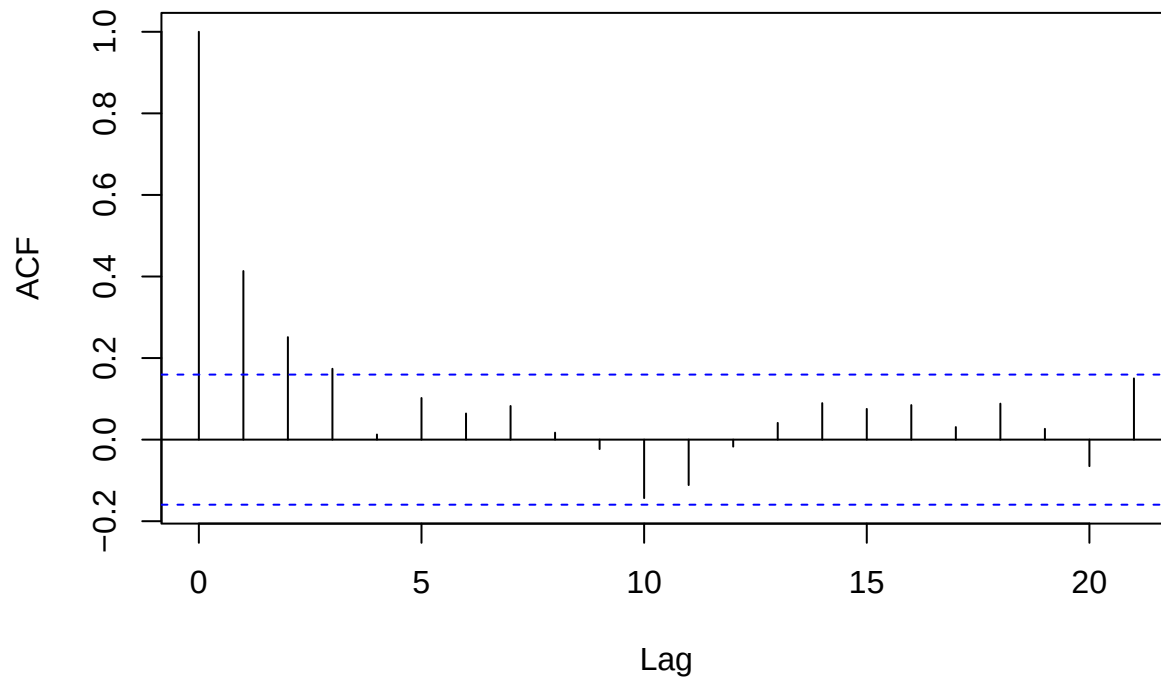
```
acf(dengue_climate_rj$tempmed, main = "ACF of Average Temperature")
```

ACF of Average Temperature



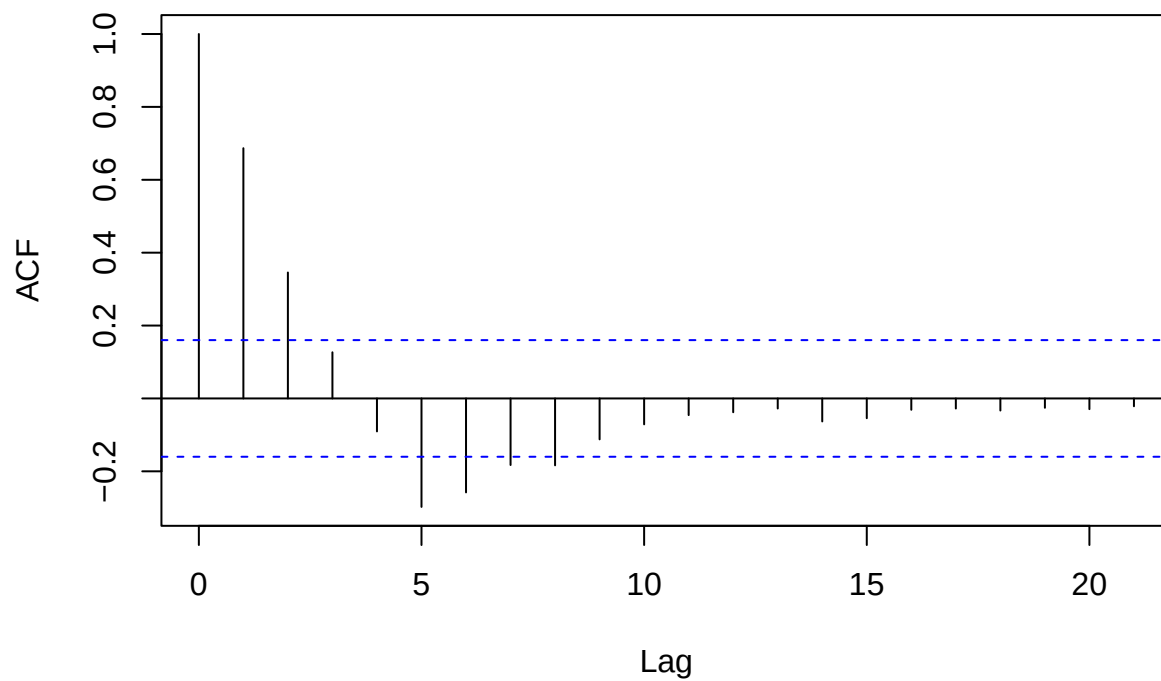
```
acf(dengue_climate_rj$umidmed, main = "ACF of Average Humidity")
```

ACF of Average Humidity

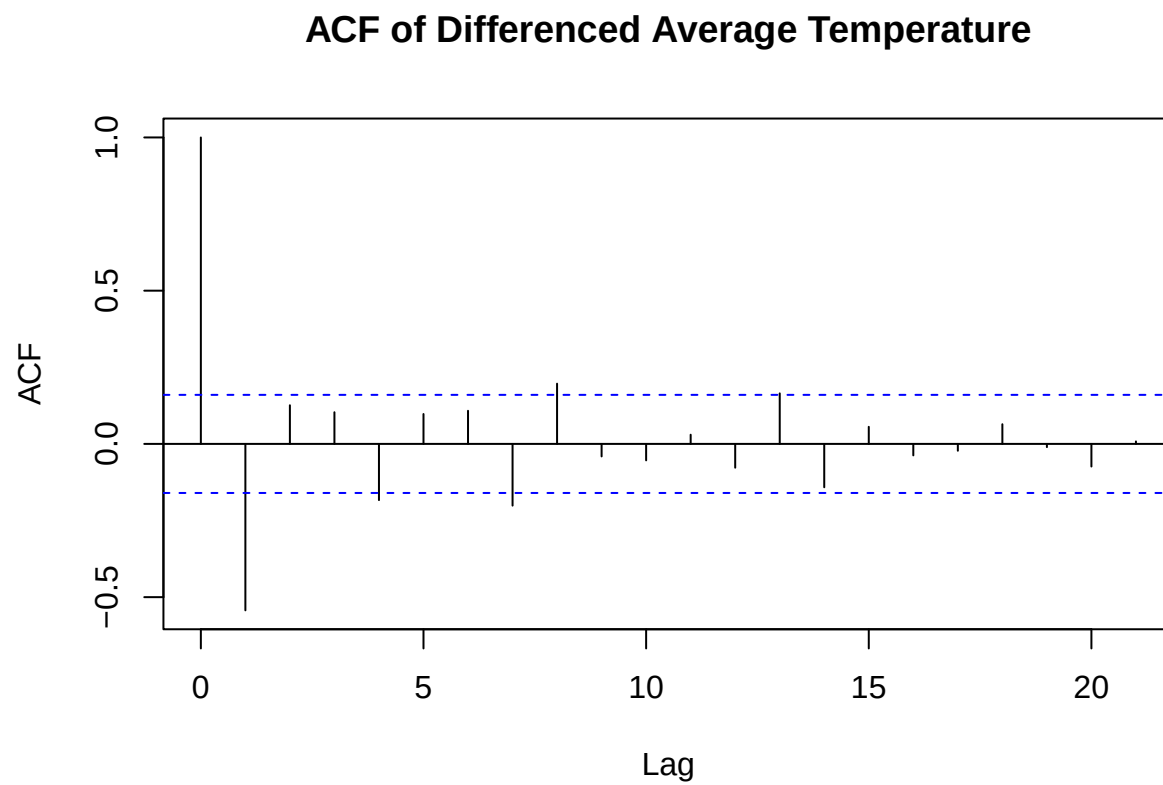


```
acf(diff(dengue_climate_rj$casprov), main = "ACF of Differenced Dengue Cases")
```

ACF of Differenced Dengue Cases



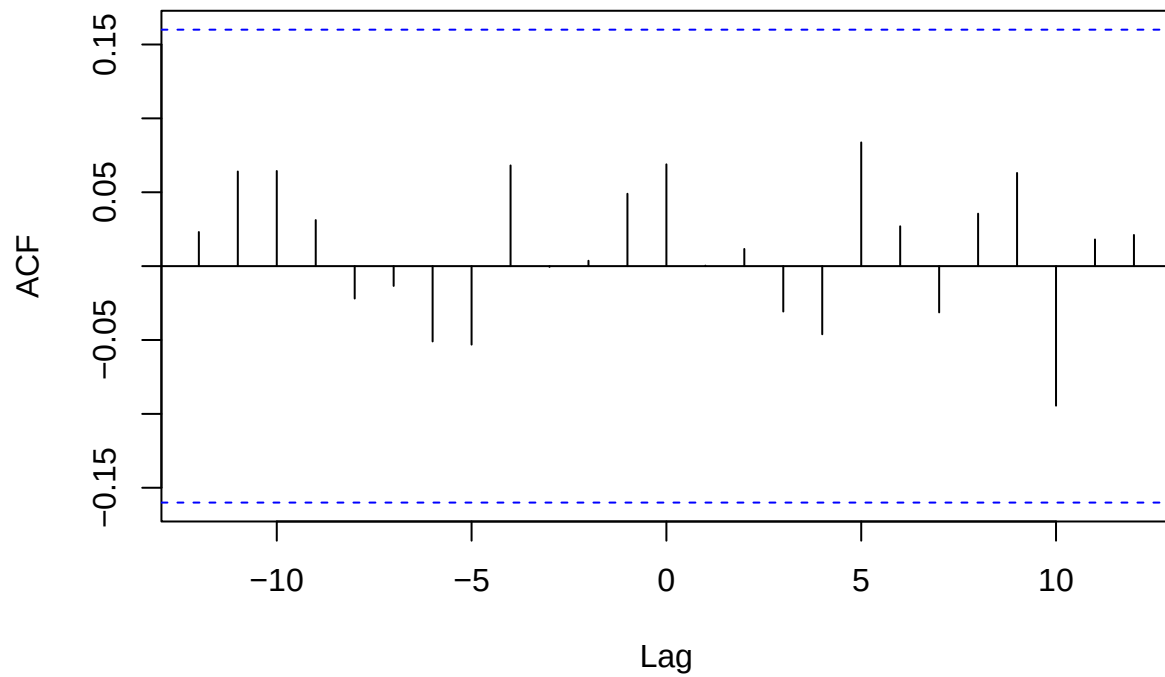
```
acf(diff(dengue_climate_rj$tempmed), main = "ACF of Differenced Average Temperature")
```



Lagged Correlation Analysis

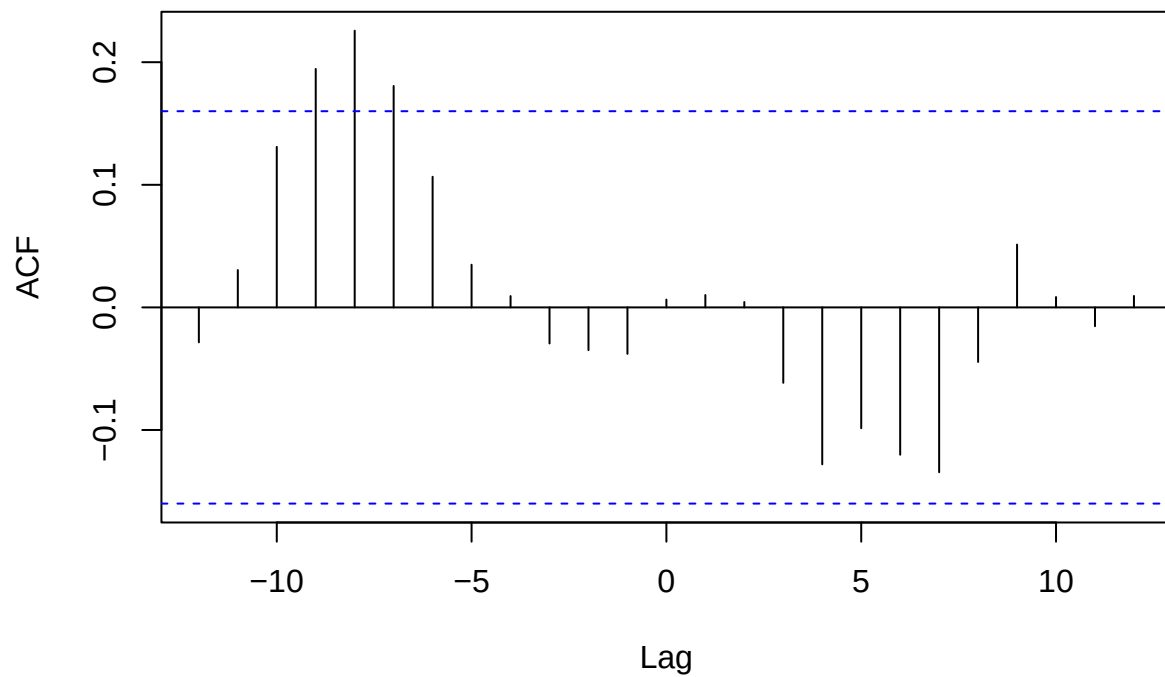
```
ccf_temp <- ccf(diff(dengue_climate_rj$casprov), diff(dengue_climate_rj$tempmed), lag.max = 12)
```

diff(dengue_climate_rj\$casprov) & diff(dengue_climate_rj\$stempmec



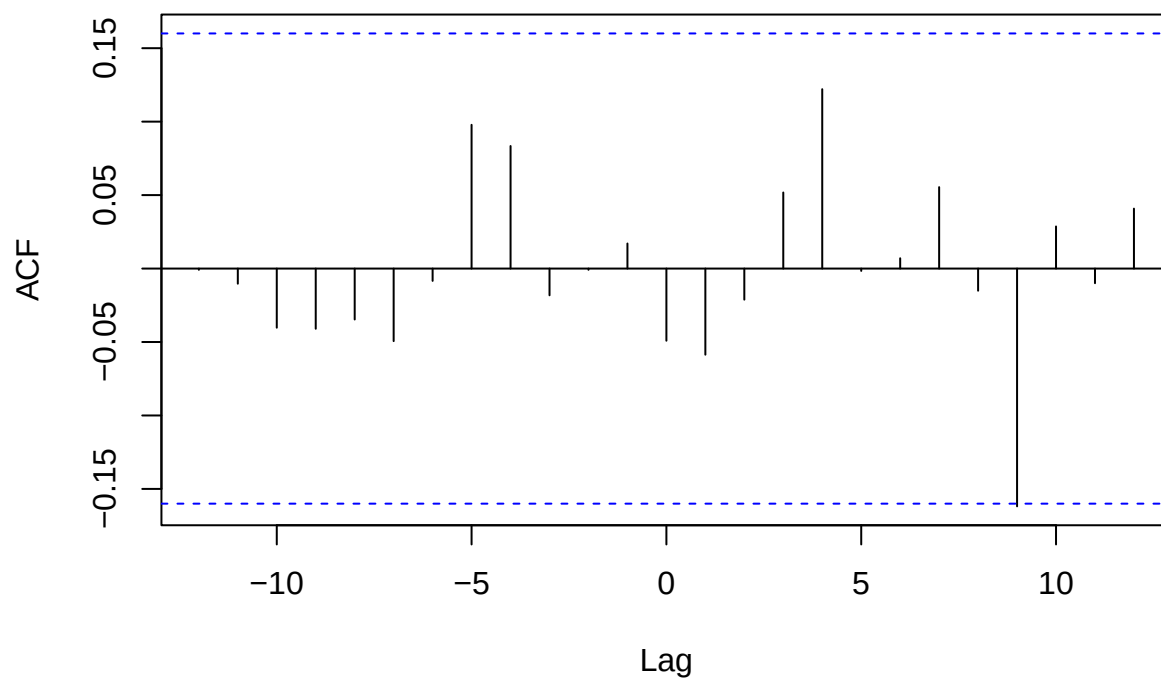
```
ccf_umid <- ccf(diff(dengue_climate_rj$casprov), dengue_climate_rj$umidmed, lag.max = 12)
```

diff(dengue_climate_rj\$casprov) & dengue_climate_rj\$umidmed



```
ccf_precip <- ccf(diff(dengue_climate_rj$casprov), diff(dengue_climate_rj$precip_total), lag.max = 12)
```

diff(dengue_climate_rj\$casprov) & diff(dengue_climate_rj\$precip_tot



Fitting Models

Model	WIS	WAIC
M1	13.36	1264.51
M2	12.62	1261.72